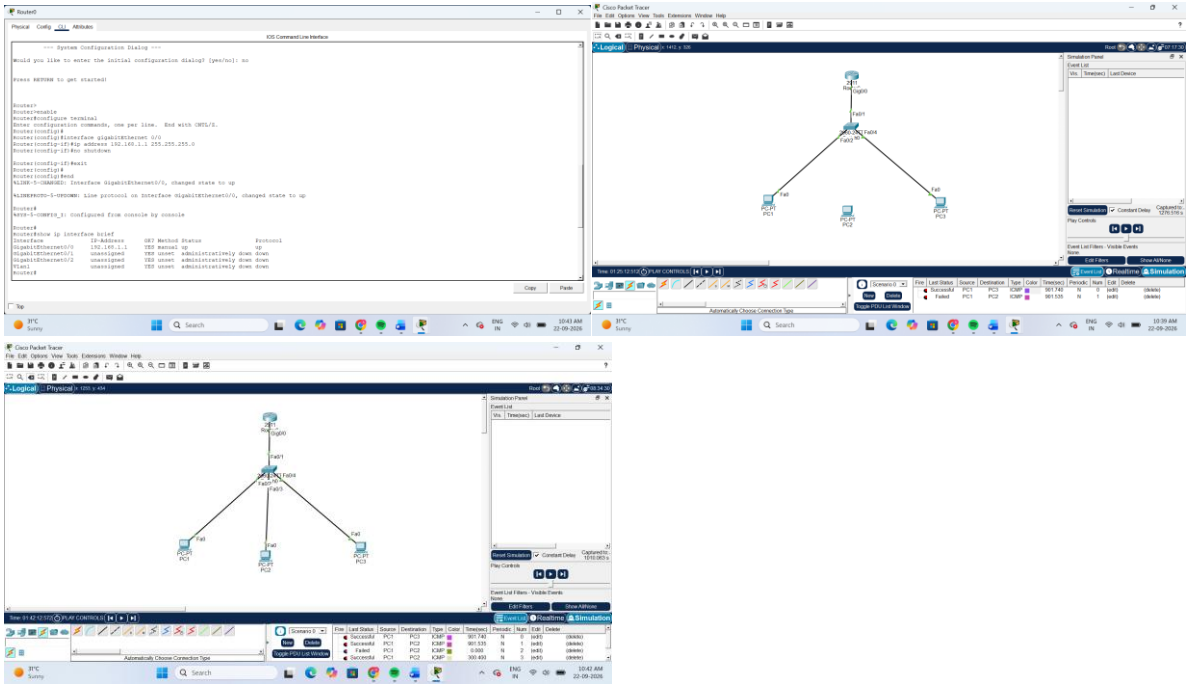
COMMON NETWORK ISSUES

1.List 5 common hardware issues that can cause network problems in a typical office setup, and for each, describe one symptom that helps identify it.

No.	Hardware Issue	Symptom
1	**Damaged Ethernet Cable**	Network connection keeps disconnecting or shows **“Network Cable Unplugged.”**
2	**Faulty Network Interface Card (NIC)**	The computer cannot connect to the network even when the cable and switch port are working.
3	**Faulty Switch Port**	Devices connected to one particular port have no network connection, while other ports work normally.
4	**Router Failure**	Multiple devices lose internet/network connectivity at the same time.
5	**Wireless Access Point Failure**	Users cannot detect or connect to the office Wi-Fi network.

2.Open Cisco Packet Tracer or any network simulator, set up a small network with 3 PCs, 1 switch, and 1 router, and intentionally disconnect one device. Use the simulator's troubleshooting tools to identify which device is faulty and document the steps you took.



Troubleshooting Steps

1. I created a small network using 3 PCs, 1 switch, and 1 router in Cisco Packet Tracer.
2. I configured the IP addresses of all three PCs and the router.
3. I first tested the network using a Simple PDU (ICMP).
4. The connection between PC1 and PC3 was successful, confirming that the network was working properly.
5. I intentionally disconnected PC2 from the switch to create a network fault.
6. I sent an ICMP packet from PC1 to PC2, which resulted in a Failed status.
7. I checked PC2's IP configuration and found it was correctly configured.
8. I then checked the physical connection and identified that PC2 was disconnected from the switch.
9. I reconnected PC2 to Switch Fa0/3 using a Copper Straight-Through cable.
10. I tested the connection again using Simple PDU.
11. The PC1 → PC2 connection was successfully restored.

Fault Identified

Fault: PC2 was physically disconnected from the switch.

Symptom: ICMP communication from PC1 to PC2 failed.

Solution: Reconnected PC2's FastEthernet0 interface to Switch Fa0/3.

**3.Create a troubleshooting checklist for diagnosing slow internet issues in a home WiFi network, including at least 8 steps covering both hardware and software checks.

Hint: Think about checking cables, router status lights, device settings, and running ping or speed tests.**

Step	Check	What to Do
1	**Check cables**	Make sure the router's power and Ethernet/Internet cables are properly connected and not damaged.
2	**Check router status lights**	Check the Power, Internet/WAN, and Wi-Fi lights. Unusual red or blinking lights may indicate a problem.
3	**Restart the router**	Turn off the router, wait about 30 seconds, and turn it back on. Wait for all required lights to become stable.
4	**Check Wi-Fi signal strength**	Move closer to the router and check whether the internet becomes faster. Weak signal can reduce speed.

5	**Check connected devices**	Disconnect unused devices because multiple devices using the same Wi-Fi can reduce available bandwidth.
6	**Check device settings**	Make sure Wi-Fi is enabled, Airplane Mode is off, and the correct Wi-Fi network is selected.
7	**Run a ping test**	Open Command Prompt and run ping 8.8.8.8 to check whether there is high delay or packet loss.
8	**Run a speed test**	Use a reliable internet speed-test website and compare the download/upload speed with the speed promised by the ISP.
9	**Check background applications**	Stop downloads, cloud synchronization, streaming, or software updates that may be consuming bandwidth.

10	**Update software/drivers**	Update the Wi-Fi adapter driver and router firmware if an update is available.

4.You receive a complaint that a user cannot access Instagram and YouTube on their laptop, but WhatsApp Web works fine. Write down the first 5 troubleshooting steps you would perform to identify if the issue is with hardware, software, or network configuration.

